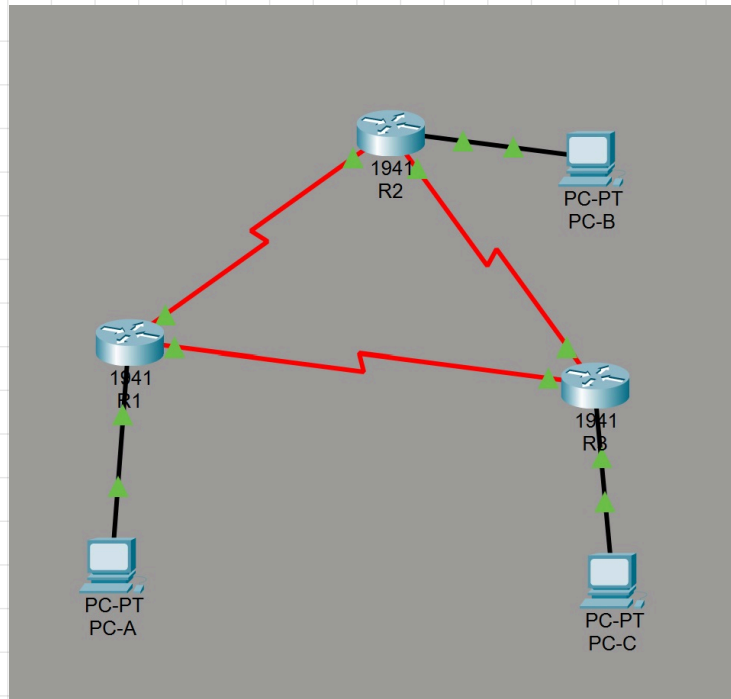


Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

show ip route ospf

```
R1#show ip route ospf
O   192.168.2.0 [110/65] via 192.168.12.2, 00:03:40, Serial0/0/0
O   192.168.3.0 [110/65] via 192.168.13.2, 00:02:42, Serial0/0/1
    192.168.23.0/30 is subnetted, 1 subnets
O       192.168.23.0 [110/128] via 192.168.12.2, 00:02:31, Serial0/0/0
                                [110/128] via 192.168.13.2, 00:02:31, Serial0/0/1
```

Part 4: Configure OSPF Passive Interfaces

- g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? Serial0/0/0

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? 129

Does R2 show up as an OSPF neighbor on R1? Yes

Does R2 show up as an OSPF neighbor on R3? No

What does this information tell you?

It shows that a passive interface still advertises its network prefix but stops sending and receiving Hello packets. Consequently, no OSPF adjacency can be formed. Setting a WAN link to passive breaks the neighbor relationship and causes OSPF to recalculate a route through other available paths.

- h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

```
R2(config-router)#router ospf 1
R2(config-router)#no passive-interface s0/0/1
R2(config-router)#
00:20:35: %OSPF-5-ADJCHG: Process 1, Nbr 33.33.33.33 on Serial0/0/1 from LOADING to FULL, Loading Done
```

- i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? Serial 0/0/1

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated? 65 R3 Serial cost (64) + R2 LAN port cost (1) = 65

Is R2 listed as an OSPF neighbor to R3? Yes

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

- a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
R1(config-router)# auto-cost reference-bandwidth 100
% OSPF: Reference bandwidth is changed.
Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

Because the default reference bandwidth is only 100 Mbps. Modern networks use Gigabit or 10-Gigabit links; increasing the reference bandwidth allows OSPF to accurately differentiate between these higher-speed interfaces by assigning them different costs.

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

The costs are calculated by dividing the reference bandwidth (100 Mbps) by the interface bandwidth (128 Kbps). For a 128 Kbps link, the cost is $100,000 / 128 = 781.25$, rounds to 781. The accumulated cost to a remote network is the sum of the exit interface costs along the path.

- i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

The new cost is 845. This is the sum of the Serial link cost (781) and the remote router's Gigabit interface cost (64).

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

The manual cost for S0/0/1 (1565) is higher than the overall cost of the path through R2. OSPF usually picks the path with the lowest total cost, thus it switched to the one that went through R2.

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

It makes sure that each router in the OSPF domain has a unique name and keeps things the same. By giving a Router ID manually, you may stop it from changing abruptly if an interface goes down or a new IP is set up. If this happens, the OSPF process would have to restart, which would make the network routing table unstable.

2. Why is the DR/BDR election process not a concern in this lab?

This lab's routers are linked using Serial interfaces that use Point-to-Point network types. Only on multi-access networks, like Ethernet, do DR/BDR elections happen. OSPF makes a full adjacency directly between the two neighbors on Point-to-Point networks.

3. Why would you want to set an OSPF interface to passive?

To make things safer and more efficient. It stops the router from delivering extra OSPF Hello packets into a LAN that doesn't have any other OSPF routers (like a network with only PCs). This saves bandwidth, lowers the stress on the CPU, and stops people who shouldn't be able to see the network topology from seeing it through OSPF packets.

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

Using Areas (like Area 0). Dividing a big network into smaller sections makes Link-State Advertisement (LSA) flooding less of a problem, shrinks the Link-State Database (LSDB), and keeps routing table calculations easy to do even as the network gets bigger.

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

network 10.0.0.0 0.255.255.255 area 0

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

The EIGRP route will be used. This is because when a router learns about the same network via multiple protocols, it looks at their Administrative Distance (AD) instead of the metric. EIGRP has the lowest AD (90), followed by OSPF (110) and RIP (120).